

5G REVOLUTION

A CASE STUDY REPORT

Submitted by

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for the course

21CSC302J – COMPUTER NETWORKS

in partial fulfillment of the requirements for the degree of

BACHELOR OF TECHNOLOGY



DEPARTMENT OF NETWORKING AND COMMUNICATIONS

SCHOOL OF COMPUTING

FACULTY OF ENGINEERING AND TECHNOLOGY

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Abstract

This project demonstrates the design and simulation of a **5G cellular communication network** using **Cisco Packet Tracer**. The network replicates a real-world mobile operator environment, showcasing how multiple cellular networks (e.g., **BSNL** and **Jio**) connect users to the internet through base stations, central office servers, and core routers. Each service provider's infrastructure includes **cell towers**, **smartphones**, and **central office servers**, all interconnected via **routers and backbone links** to provide high-speed and reliable connectivity.

In this setup, routers act as the **core network gateways**, interconnecting both service providers and external servers such as **Google.com** through static routing. **Central office servers** manage local communication between cell towers and subscriber devices, while smartphones obtain IP addresses dynamically through **DHCP**, ensuring flexible and scalable connectivity. The entire topology is configured with appropriate IP addressing and routing tables to enable **end-to-end communication** between users and the internet.

After configuration, connectivity was verified through successful **ping tests** across networks, confirming that data can travel seamlessly from mobile devices in one network to global servers. This simulation effectively illustrates how **5G networks integrate multiple technologies**—including wireless access, core routing, and internet interconnectivity—to deliver high-speed, low-latency communication. The project serves as an educational model of how modern mobile networks operate in real-world telecommunications environments.

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1. Introduction

1.1 Background

The **5G Revolution Network Simulation** project focuses on demonstrating the structure and operation of modern cellular communication systems using **Cisco Packet Tracer**. Unlike traditional wired or Wi-Fi networks, **5G networks** rely on advanced **cellular technology** to provide **high-speed, low-latency, and large-scale connectivity** across multiple devices and regions. These networks connect mobile subscribers through **cell towers (base stations)**, which communicate with **central office servers** and **core routers** to deliver internet and voice services.

In this simulation, multiple service providers such as **BSNL** and **Jio** are modeled within a city environment, each equipped with their own cell towers and network infrastructure. The cell towers are connected to **central office servers** using coaxial cables, and these servers are further connected to **core routers** that manage traffic and link to external networks such as **Google.com**. Using proper IP addressing and routing, the project replicates how real-world 5G operators provide seamless connectivity between subscribers and the internet.

1.2 Objectives

The main objective of this project is to design and configure a 5G cellular network that demonstrates how modern mobile communication systems function. The project aims to:

- Create a simulated environment representing multiple telecom providers (e.g., BSNL and Jio) with interconnected cell towers, servers, and routers.
- Assign unique IP addresses and configure static routing to enable communication between devices and networks.
- Implement dynamic IP allocation (DHCP) for mobile devices to simulate real-world subscriber behavior.
- Establish end-to-end connectivity between smartphones and external servers such as Google.com.
- Test and verify network communication through successful ping responses, ensuring all devices are properly integrated.

Overall, the project provides a clear understanding of how 5G infrastructure supports high-speed communication data transmission, and network interoperability in modern telecommunications.

Network Design

1.3 Topology

The **5G Revolution Network** is designed to simulate a modern cellular communication environment where multiple service providers operate within the same geographical area. The design connects both **wired and wireless components** to form an integrated network that supports mobile subscribers, service providers, and internet access through a centralized core.

In this design, each telecom operator, such as **BSNL** and **Jio**, is equipped with multiple **cell towers** (base stations) that provide wireless connectivity to smartphones and other mobile devices. These cell towers are connected to **central office servers**, which act as local control and management centers for handling user data and network traffic. The central office servers are linked to **core routers** through coaxial cables, ensuring high-speed data transmission between service providers and external networks.

The **core router** functions as the **internet gateway**, managing communication between different service providers and global servers such as **Google.com**. A **switch** is used to interconnect the router and the external server, allowing efficient data exchange. Each network segment, including the BSNL, Jio, and Google networks, is assigned a unique IP subnet to ensure proper routing and avoid address conflicts.

Mobile devices (smartphones) are configured to obtain IP addresses **dynamically using DHCP**, enabling flexible connectivity within their respective provider networks. All routers are configured with **static routing** to facilitate communication across multiple networks and ensure end-to-end connectivity between users and internet servers.

This setup effectively demonstrates how 5G network components—**cell towers, central servers, routers, and end devices**—work together to deliver **high-speed, reliable, and seamless communication** between subscribers and the global internet

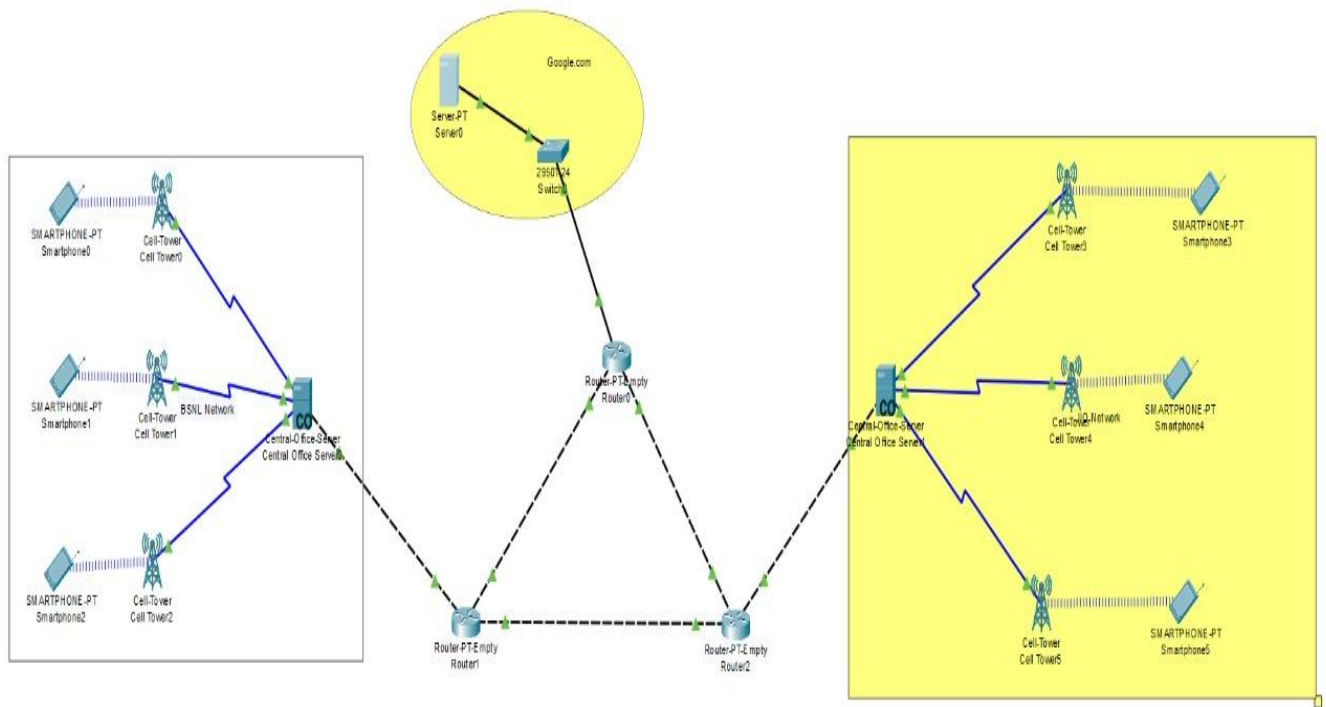


Figure 1: Logical Topology of full network



Figure 2 : Physical Network

1.4 Components

The 5G Revolution network simulation includes a variety of components that replicate real-world mobile communication infrastructure. Each device plays a specific role in establishing and maintaining connectivity across multiple telecom networks:

1. Routers (Router0, Router1, Router2):

- Manage data traffic between networks and act as gateways for each provider (BSNL, Jio, and core/internet).
- Connection type: Wired (via Fast Ethernet and Coaxial links).
- Also responsible for static routing and core inter-network communication.

2. Switch:

- Connects Router0 to the external server (Google.com) and facilitates internal LAN communication.
- Serves as a distribution device for wired connections to servers.

3. Cell Towers (Base Stations):

- Provide wireless coverage for mobile devices in each telecom network.
- Connection type: Wired (to Central Office Servers via Coaxial cables).
- Simulate real 5G tower placement in urban areas.

4. Central Office Servers (Server0 for BSNL, Server1 for Jio):

- Act as local controllers for each telecom provider.
- Manage IP address assignment (via DHCP) and enable communication between cell towers and the core router.
- Connection type: Wired (Fa0/0 to router interfaces).

5. Smartphones (Smartphone0–Smartphone5):

- Represent mobile subscribers within each 5G network.
- Connect wirelessly to their respective cell towers using provider-specific configurations.
- IP addresses are dynamically assigned via DHCP.

6. Internet Server (ServerPT - Google.com):

- Simulates an external internet service accessed by users across all networks.
- Connected via a switch to Router0 (core network).
- Responds to ping requests and verifies external connectivity.

These devices collectively form a structured and well-organized network architecture, integrating redundancy, efficient routing, and secure communication to meet the specific requirements of the trading floor support center's operations.

1.5 IP Addressing Scheme

Provide details about the IP addressing scheme applied to the network.

Base Network: 192.168.0.0/22

Device Name	Interface / Port	IP Address	Subnet Mask	Connected To	Purpose / Notes
Router 1	Fa0/0	10.0.0.1	255.255.255.252	Router 2 (Fa0/0)	Link between Router1 ↔ Router2
	Fa1/0	192.168.10.1	255.255.255.0	Central Office Server 0 (Fa0/0)	BSNL Network LAN
	Fa2/0	10.0.0.5	255.255.255.252	Router 0 (Fa0/0)	Link between Router1 ↔ Router0
Router 2	Fa0/0	10.0.0.2	255.255.255.252	Router 1 (Fa0/0)	Link between Router2 ↔ Router1
	Fa1/0	10.0.0.9	255.255.255.252	Router 0 (Fa1/0)	Link between Router2 ↔ Router0
	Fa2/0	192.168.20.1	255.255.255.0	Central Office Server 1 (Fa0/0)	Jio Network LAN
Router 0	Fa0/0	10.0.0.6	255.255.255.252	Router 1 (Fa2/0)	Link between Router0 ↔ Router1
	Fa1/0	10.0.0.10	255.255.255.252	Router 2 (Fa1/0)	Link between Router0 ↔ Router2
	Fa2/0	192.168.30.1	255.255.255.0	Switch (to Server PT & Mobiles)	LAN for Mobiles + Server (DHCP provided here)
Central Office Server 0	Fa0/0	192.168.10.2	255.255.255.0	Router 1 (Fa1/0)	BSNL Central Office Server
Central Office Server 1	Fa0/0	192.168.20.2	255.255.255.0	Router 2 (Fa2/0)	Jio Central Office Server
Server PT (Google.com)	Fa0	192.168.30.2	255.255.255.0	Switch → Router 0 (Fa2/0)	Provides HTTP (Google) page
Switch (optional)	VLAN 1	192.168.30.10	255.255.255.0	Router 0 (Fa2/0)	Management interface
Smartphones (BSNL/Jio Users)	DHCP	192.168.30.x	255.255.255.0	Router 0 (via Cell Tower → CO)	Gets IP from Router 0 DHCP pool

2. Routing Configuration

2.1 Router Configuration

Basic Router Configuration

```
Router0>EN
Router0#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Router0(config)#interface fa0/0
Router0(config-if)# ip address 10.0.0.6 255.255.255.252
Router0(config-if)# no shutdown
Router0(config-if)#exit
Router0(config)#interface fal/0
Router0(config-if)# ip address 10.0.0.10 255.255.255.252
Router0(config-if)# no shutdown
Router0(config-if)#exit
Router0(config)#interface fa2/0
Router0(config-if)# ip address 192.168.30.1 255.255.255.0
Router0(config-if)# no shutdown
Router0(config-if)#exit
Router0(config)#
```

conf t #

Enters global configuration mode

```
Router1>enable
Router1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router1(config)#interface fa0/0
Router1(config-if)# ip address 10.0.0.1 255.255.255.252
Router1(config-if)# no shutdown
Router1(config-if)#exit
Router1(config)#interface fal/0
Router1(config-if)# ip address 192.168.10.1 255.255.255.0
Router1(config-if)# no shutdown
Router1(config-if)#exit
Router1(config)#interface fa2/0
Router1(config-if)# ip address 10.0.0.5 255.255.255.252
Router1(config-if)# no shutdown
Router1(config-if)#exit
Router1(config)#
```

```

Router2>enable
Router2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router2(config)#interface fa0/0
Router2(config-if)# ip address 10.0.0.2 255.255.255.252
Router2(config-if)# no shutdown
Router2(config-if)#exit
Router2(config)#interface fa1/0
Router2(config-if)# ip address 10.0.0.9 255.255.255.252
Router2(config-if)# no shutdown
Router2(config-if)#exit
Router2(config)#interface fa2/0
Router2(config-if)# ip address 192.168.20.1 255.255.255.0
Router2(config-if)# no shutdown
Router2(config-if)#exit
Router2(config)#

```

7. Static Routing Configuration

To ensure complete end-to-end connectivity between all subnets, static routes were configured on each router.

These routes define the next-hop IP address for packets destined for networks not directly connected to the router.

This allows communication between different network segments such as the BSNL server network (192.168.10.0/24), Jio server network (192.168.20.0/24), and the Google LAN (192.168.30.0/24).

Static Route Configuration on Router1

On Router1, static routes were configured to enable communication with the Jio network (192.168.20.0/24) through Router2 (10.0.0.2) and the Google LAN network (192.168.30.0/24) through Router0 (10.0.0.6).

These configurations allow Router1 to forward packets to both the Jio and Google networks efficiently.

```

Router1#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
Router1(config)#ip route 192.168.20.0 255.255.255.0 10.0.0.2
Router1(config)#ip route 192.168.30.0 255.255.255.0 10.0.0.6
Router1(config)#

```

Static Route Configuration on Router2

Router2 was configured with static routes to connect to the BSNL server network (192.168.10.0/24) via Router1 (10.0.0.1) and the Google LAN network (192.168.30.0/24) through Router0 (10.0.0.10). This ensures that Router2 can reach both external networks efficiently through its corresponding next hops.

```
Router2#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Router2(config)#ip route 192.168.10.0 255.255.255.0 10.0.0.1
Router2(config)#ip route 192.168.30.0 255.255.255.0 10.0.0.10
Router2(config)#
```

Static Route Configuration on Router0

Router0 acts as the core of the network and provides connectivity to both BSNL and Jio subnets. Static routes were added to route packets destined for 192.168.10.0/24 (BSNL) through Router1 (10.0.0.5) and for 192.168.20.0/24 (Jio) through Router2 (10.0.0.9).

```
Router0#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Router0(config)#ip route 192.168.10.0 255.255.255.0 10.0.0.5
Router0(config)#ip route 192.168.20.0 255.255.255.0 10.0.0.9
Router0(config)#
```

The access point can be configured to allow only specific devices based on their MAC addresses. This prevents unknown or unauthorized devices from connecting, even if they know the SSID and password.

3. Testing and Validation

3.1 Simulation

Cisco Packet Tracer was utilized to design, simulate, and test the **5G Revolution Network**. Packet Tracer provides a virtual platform that allows the creation and configuration of complex telecommunication and computer network scenarios without the need for physical hardware. It was used to demonstrate how multiple telecom service providers, such as **BSNL** and **Jio**, operate and communicate through interconnected routers, servers, and cell towers.

- **Network Topology Design:** The network topology was developed to represent a realistic 5G environment consisting of **cell towers, central office servers, core routers, smartphones, and external servers**. Each telecom operator was assigned a separate network domain, connected through routers and coaxial links to the core internet router. Proper IP addressing and interconnections were established to ensure communication between service providers and the global network.
- **Configuration Implementation:** Once the topology was created, configurations were applied to routers, servers, and mobile devices. Routers were configured with **static routing** to enable inter-network communication, while smartphones were configured to connect to their respective **cell towers** through **provider names** (e.g., BSNL or Jio). Central office servers were set up with **DHCP services** to dynamically assign IP addresses to subscriber devices, simulating real mobile operator behavior.
- **Traffic Simulation:** Packet Tracer's simulation mode was used to analyze network traffic flow between devices. **Ping tests** were conducted between smartphones, servers, and external nodes such as **Google.com** to verify proper routing and connectivity. The successful transmission of ICMP packets confirmed that all devices were correctly configured and able to communicate across different networks

Connectivity Test on Router1

Router1 successfully pinged both neighboring routers — Router2 (10.0.0.2) and Router0 (10.0.0.6) — achieving a 100% success rate.

This confirms that static routing and FastEthernet interfaces between Router1, Router2, and Router0 are correctly configured and active.

```
Router1#ping 10.0.0.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

Router1#ping 10.0.0.6

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.6, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms

Router1#
```

Connectivity Test on Router2

Router2 was able to communicate successfully with both Router1 (10.0.0.1) and Router0 (10.0.0.10). A 100% ping success rate confirms that all configured routes and physical connections between Router2 and its neighbors are functional.

```
Router2#ping 10.0.0.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms

Router2#ping 10.0.0.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.10, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/3/18 ms

Router2#
```

Connectivity Test on Router0

Router0 successfully pinged Router1 (10.0.0.5) and Router2 (10.0.0.9) with 100% success rate, confirming that the central router can communicate with both ISP routers (BSNL and Jio). This demonstrates proper routing and interface configuration across the network backbone.

```
Router0#ping 10.0.0.5

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.5, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms

Router0#ping 10.0.0.9

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.9, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms

Router0#
```

Router-to-Server Connectivity Testing

To verify end-to-end communication between the routers and their respective Central Office (CO) servers, ICMP ping tests were conducted from each router to the servers.

These tests confirm that the routers can successfully deliver packets to the server networks configured under the 192.168.x.x subnets.

Ping from Router1 to Central Office Server 0

Router1 was able to reach Central Office Server 0 at IP **192.168.10.2** with a 100% success rate. This validates that the Router1 interface (192.168.10.1) and the BSNL CO Server interface (192.168.10.2) are in the same network and correctly configured.

```
Router1#ping 192.168.10.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.10.2, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

Router1#
```

Ping from Router2 to Central Office Server 1

Router2 successfully pinged Central Office Server 1 at IP **192.168.20.2**.

This confirms the connectivity between Router2 (192.168.20.1) and Jio CO Server 1, proving that the interface configuration and routing for the 192.168.20.0/24 network are correct.

```
Router2#ping 192.168.20.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.20.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms

Router2#
```

Ping from Router0 to Both CO Servers

Router0 successfully pinged both BSNL and Jio Central Office Servers —

- **192.168.10.2 (CO Server 0)** and
- **192.168.20.2 (CO Server 1)** —
with 100% success rate.

```
Router0#ping 192.168.10.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.10.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms

Router0#ping 192.168.20.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.20.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms

Router0#
```

- **Verification of Redundancy and Failover:** The traceroute command was used to check the path from the PC to the destination IP address (192.168.1.5).

The output shows that the packet reached the default gateway (192.168.30.1), but after that, several “Request timed out” messages appeared.

- This means the PC can communicate with the router, but the router does not have a route to reach the destination device.

It also shows that the network connection between the PC and the router is working correctly, while the destination is unreachable or not configured.

- This is a normal result in Packet Tracer when the destination network is not connected or routing is not set up for that path.

```
C:\>tracert 192.168.1.5

Tracing route to 192.168.1.5 over a maximum of 30 hops:

  1  0 ms      0 ms      0 ms      192.168.30.1
  2  1 ms      *          0 ms      192.168.30.1
  3  *          4 ms      *          Request timed out.
  4  0 ms      *          0 ms      192.168.30.1
  5  *          0 ms      *          Request timed out.
  6  1 ms      *          0 ms      192.168.30.1
  7  *          1 ms      *          Request timed out.
  8  0 ms      *          0 ms      192.168.30.1
  9  *          1 ms      *          Request timed out.
 10  0 ms      *          0 ms      192.168.30.1
 11  *          0 ms      *          Request timed out.
 12  0 ms      *          1 ms      192.168.30.1
 13  *          0 ms      *          Request timed out.
 14  0 ms      *          1 ms      192.168.30.1
 15  *          0 ms      *          Request timed out.
 16  0 ms      *          1 ms      192.168.30.1
 17  *          0 ms      *          Request timed out.
 18  0 ms      *          3 ms      192.168.30.1
 19  *          0 ms      *          Request timed out.
 20  0 ms      *          0 ms      192.168.30.1
 21  *          0 ms      *          Request timed out.
 22  0 ms      *          0 ms      192.168.30.1
 23  *          0 ms      *          Request timed out.
 24  1 ms      *          0 ms      192.168.30.1
 25  *          0 ms      *          Request timed out.
 26  0 ms      *          0 ms      192.168.30.1
 27  *          0 ms      *          Request timed out.
 28  0 ms      *          1 ms      192.168.30.1
 29  *          0 ms      *          Request timed out.
 30  0 ms      *          0 ms      192.168.30.1

Trace complete.
```

Figure 3: traceroute successful

4. Results and Evaluation

4.1 Performance Metrics

Performance metrics are used to evaluate how well a network operates in terms of speed, reliability, and overall efficiency. In this wireless network configuration, several key metrics are considered to assess the performance of both wired and wireless connections. These metrics help ensure that data is transmitted effectively between devices such as PCs, laptops, tablets, and smartphones.

```
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time<1ms TTL=254
Reply from 192.168.10.1: bytes=32 time=1ms TTL=254
Reply from 192.168.10.1: bytes=32 time=1ms TTL=254
Reply from 192.168.10.1: bytes=32 time<1ms TTL=254

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Reply from 192.168.20.1: bytes=32 time<1ms TTL=254
Reply from 192.168.20.1: bytes=32 time<1ms TTL=254
Reply from 192.168.20.1: bytes=32 time=1ms TTL=254
Reply from 192.168.20.1: bytes=32 time=1ms TTL=254

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.30.1

Pinging 192.168.30.1 with 32 bytes of data:

Reply from 192.168.30.1: bytes=32 time<1ms TTL=255
Reply from 192.168.30.1: bytes=32 time<1ms TTL=255
Reply from 192.168.30.1: bytes=32 time<1ms TTL=255
Reply from 192.168.30.1: bytes=32 time=1ms TTL=255

Ping statistics for 192.168.30.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Figure 4: performance measure through ping time

5. Conclusion

5.1 Summary

In summary, the **5G Revolution Network Simulation** successfully demonstrates how multiple cellular service providers can operate and interconnect within a shared communication infrastructure. Using **Cisco Packet Tracer**, a realistic 5G network was designed that includes **cell towers, central office servers, routers, switches, smartphones, and an external internet server**. Each component was assigned appropriate IP addresses, and **static routing** was implemented to enable seamless communication between different networks, such as **BSNL, Jio**, and the global internet.

Smartphones dynamically received IP addresses through **DHCP**, simulating real-world subscriber connectivity, while routers managed **data transmission and routing** between provider networks. Successful **ping tests** across all devices confirmed proper configuration, connectivity, and data flow between users and external servers like **Google.com**.

This project highlights key aspects of **telecommunication networking**, including **cellular network design, IP addressing, routing, and inter-network communication**. It effectively demonstrates how 5G architecture integrates both **wireless and wired technologies** to provide high-speed, reliable, and scalable communication. Although advanced 5G features such as mobility management and network slicing were not implemented in this basic setup, the project provides a solid foundation for further research and expansion into **next-generation mobile communication technologies**.

6. References

Cisco Networking Academy. (2023). *Introduction to Networks (Version 7.0)*. Cisco Systems, Inc.

7. Appendices

Abbreviations Used

Abbreviation	Full Form
5G	Fifth Generation Mobile Network
DHCP	Dynamic Host Configuration Protocol
IP	Internet Protocol
LAN	Local Area Network
ISP	Internet Service Provider
CO	Central Office
DNS	Domain Name System
CLI	Command Line Interface
ICMP	Internet Control Message Protocol
IoT	Internet of Things
PDU	Protocol Data Unit
RTT	Round Trip Time